

Jiayu Chen

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Applied ML researcher developing physiological sensing systems, with strengths in rapid prototyping, large-scale clinical data collection and management, real-time deployment, and FDA-cleared product development.

EDUCATION

University of Georgia | Athens, GA | Aug 2024 - Present (Estimated 2028)

Ph.D. in Electrical and Computer Engineering | GPA: 4.0/4.0 | Advisor: Prof. Wenzhan Song

Hohai University | Nanjing, China | Sep 2020 - May 2024

B.Eng. in Software Engineering | GPA: 4.80/5.00

RESEARCH EXPERIENCE

University of Georgia | Ph.D. | Athens, GA | Aug 2024 - Present

- **Respiratory & Sleep Monitoring:** Developed the core contactless respiratory-rate (RR) algorithm for FDA-cleared BedDot (G1). Developed SeismoApnea, a contact-free tri-axial geophone system for overnight sleep-apnea monitoring that estimates total sleep time (TST), Apnea-Hypopnea Index (AHI), and four-class apnea severity. Across 80 overnight recordings from 80 subjects, it achieved an AHI ICC of 0.870 and a severity macro F1 of 73.11%. FDA clearance is being pursued specifically for AHI estimation.
- **Cardiovascular Monitoring:** Developed and deployed SelfDenoiser, a self-supervised temporal-spectral framework trained on 11,392 hours of ICU BCG recordings, as the core denoising module for contactless cardiac monitoring. The module improves downstream heart-rate and inter-beat-interval estimation and is part of a filed FDA submission.
- **Office Health Monitoring:** Developed a contactless vital-sign monitoring system for real-world office environments, achieving heart-rate and respiratory-rate MAEs of approximately 3.9 bpm and 2.4 breaths/min, with 88% and 21% temporal coverage, respectively. Following an on-site visit led by a MillerKnoll Senior Vice President, the company procured 30 units for pilot testing.
- **Infant Monitoring:** Data collection is underway at Children's Healthcare of Atlanta (CHOA) using a contactless tri-axis geophone system in bassinet, incubator, and crib settings. Preliminary signal-quality analysis demonstrates the feasibility of monitoring occupancy, heart rate, respiratory rate, sleep/wake states, and crying.

SELECTED IMPACTS

- Developed the core contactless respiration-monitoring algorithm for BedDot (G1), contributing to FDA 510(k) clearance K253274 as a breathing-frequency monitor (June 2026). FDA validation covered 52 adults and reported a respiratory-rate MAE of 1.77 breaths/min.

SELECTED PUBLICATIONS

- J. Chen, et al. *SelfDenoiser: Self-supervised Seismic Signal Denoiser for Continuous and Contactless Cardiac Monitoring*. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (PACM IMWUT)*, 9(4), 2025. [CORE A* | CCF-A]
- Y. Song*, J. Chen*, et al. *Contactless Sleep Apnea Detection with Bodyseismography*. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (PACM IMWUT)*, 10(2), 2026. [CORE A* | CCF-A]
- Y. Song*, J. Chen*, et al. *Attention Feature Fusion with Cluster Contrastive Learning for Snoring and Breath-Holding Detection Using Seismic Sensing*. *IEEE International Conference on Pervasive Computing and Communications (PerCom)*, 2026. [CORE A* | CCF-B]
- Y. Song*, J. Chen*, et al. *SeatVitals: Geophone-Based Unobtrusive Vital Sign Monitoring for Office Environments*. Accepted, *Computing in Cardiology (CinC)*, 2026.

* Equal contribution.

TECHNICAL SKILLS

Methods: Self-Supervised Learning, Contrastive Learning, Time-Series Modeling, Signal Processing

Frameworks: PyTorch, scikit-learn, NumPy, SciPy, pandas

Signals: BCG/BSG, ECG, PPG

Systems: Real-Time Inference, MQTT, InfluxDB, Grafana, Linux

Programming: Python, C, SQL, LaTeX, Git

HONORS AND SERVICE

Honor: National Scholarship of China

Peer Review: ACM IMWUT/UbiComp and IEEE Journal of Biomedical and Health Informatics (JBHI)

Teaching: Teaching Assistant, CSEE 8300: Principles of Cyber-Physical Systems